Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/21/2023
PI Name:	Douglas, Grace Ph.D.		
Project Title:	Food Acceptability, Menu Fatigue, and Aversion on ISS Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	grace.l.douglas@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	NASA Johnson Space Center		
PI Address 1:	2101 NASA Parkway, Mail Code SF411		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/25/2017	**End Date:**	09/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Young, Millennia Ph.D. (NASA Johnson Space Center) Bell, Suzanne Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
	The acceptability of the food system has been linked to caloric intake and associated nutritional benefits. The diets of the crewmembers during a mission are restricted to 200 processed and prepackaged United States On-orbit Segment (USOS) standard menu items supplemented with limited preference foods. International Space Station (ISS) crew members have noted in debriefs that they would prefer more food variety for the length of the missions and they tire of certain foods over six months. Recent bed-rest data has indicated that repeat consumption of 215 foods over 70 days resulted in decreased liking of most foods, some by nearly 50% (Hunter and Caldwell, preliminary data). It is possible that menu fatigue leads to acceptability decreases and aversion of available foods that may contribute to the body mass loss often experienced by ISS crew. However, the impact of repeat food consumption on acceptability within the current spaceflight food system has not yet been investigated. Limited variety and crew preferences within food categories may have more severe health consequences as mission		

Task Description:	length increases. Diet diversity has been linked to lower incidences of gastric cancer, brain health, psychological function, and gut health (Foster and McVey Neufeld 2013; Heiman and Greenway 2016; Shiraseb et al. 2016; Vecchia et al. 1997). The diet on a long duration mission must contain adequate diversity to prevent menu fatigue and food aversion and support continued intake of a variety of nutritional foods that promote crew health. Before we can define adequate diversity for multi-year missions we need to determine the current state of acceptability of the ISS food system throughout 6-month and 1-year missions. Specific Aims: Hypothesis: Characterizing the relationship between food acceptability versus repeatability will contribute to defining requirements for an acceptable food system that will support crew health and performance on long duration missions. Aims: 1. Determine the impact of repeat consumption on food acceptability on ISS within the current closed-variety spaceflight food system. 2. Inform strategies to improve food system composition. References: Foster JA, McVey Neufeld KA. Gut-brain axis: how the microbiome influences anxiety and depression. Trends Neurosci. 2013 May;36(5):305-12. https:// . PMID: 23384445 . Heiman ML, Greenway FL. A healthy gastrointestinal microbiome is dependent on dietary diversity. Mol Metab. 2016 Mar 5;5(5):317-20. https:// . PMID: 27110483 ; PMCID: PMC4837298 . Shiraseb F, Siassi F, Qorbani M, Sotoudeh G, Rostami R, Narmaki E, Yavari P, Aghasi M, Shaibu OM. Higher dietary diversity is related to better visual and auditory sustained attention. Br J Nutr. 2016 Apr;115(8):1470-80. https:// . PMID: 26902532 . La Vecchia C, Muñoz SE, Braga C, Fernandez E, Decarli A. Diet diversity and gastric cancer. Int J Cancer. 1997 Jul 17;72(2):255-7. https:// ;255::aid-ijc9>3.0.co;2-q . PMID: 9219829 .
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. The reasons that this is a directed study are: 1) The variety and volume of the food have an impact on the design of the deep space mission habitat, for which we have a highly constrained time schedule to provide inputs. 2) This proposal is for a highly constrained pre/in/postflight data collection process, involving operational practices and personnel that are very specific to NASA. It will be implemented under the auspices of NASA's Human Research Program using existing in-house infrastructure and expertise. The study is a cross-element collaboration between Human Health Countermeasures (HHC) and Human Factors and Behavioral Performance (HFBP). In order to target current and upcoming one year missions and increase the amount of subjective data that may be collected prior to an upcoming ground based analog study we propose this as a low cost directed data assessment. The acceptability questionnaire has already been developed by the Space Food Systems Laboratory and will be similar to what crewmembers use to score the food system in pre-flight food tasting sessions. The Space Food Systems Laboratory personnel has immediate access to unique operational metadata, such as lists of crew specific menu items that were launched to ISS, real-time monitoring of open containers in flight, pre-flight operational tasting session scores, and familiarity with the ISS Food Intake Tracker (FIT) app database for ease of data handling, quick integration and processing, keeping this study low-cost. As a directed study, this questionnaire can be quickly transferred to spaceflight to take advantage of the current extended duration mission to better inform future long duration scenarios.
Research Impact/Earth Benefits:	Strategies that may improve consumption in a closed spaceflight food system could be used for closed food systems on Earth.
Task Progress:	JULY 2023 UPDATE (Ed., 7/18/23) We collected data from 15 astronauts on the International Space Station (ISS) to determine the impact of repeat consumption on food acceptability within the current closed-variety spaceflight food system (aim 1) and inform strategies to improve food system composition (aim 2). Data suggested that astronauts limited selections to favorites early in the mission and continued to like those foods throughout the mission and score them acceptably. Data suggested the importance of variety and choice: many crewmembers commented on the lack of variety in foods, particularly with vegetables and breakfast items, and the importance of augmenting menu fatigue with supplemental items from crew specific menus, international partner foods and other ‘enhancements’. These results can inform strategies to improve the food system for current ISS operations, but also suggests knowledge and intervention gaps for future exploration missions. Although up-mass and food lab limitations limit the amount of variety that can be provided in the standard menu, the comments and scores from this study and diet records from the ISS have already been used to start updating the ISS menu. Specifically, updates have aimed to provide more fruits and vegetables, increase breakfast proteins, and reduce carbohydrates. It is important to continue to obtain feedback like the data here from crewmembers as these menu changes are implemented to identify evolution in general dietary patterns and acceptability of the changes instituted. Some future menu plans for highly restricted resource missions limit astronauts to only the calories that they need, despite previous indication that if crewmembers do not want to consume an item they will lose weight rather than consume it. In addition, food may be prepositioned, eliminating the potential to provide preference foods, as this would not support late crew assignment changes. Data from this study, and future food acceptability ratings of new menu items, can be used to help design menus that are most acceptable for those crews. At the same time, many of the strategies used by astronauts to increase variety and ward off menu fatigue, such as crew specific items, are likely to be limited or not available for exploration missions. This requires additional work to understand how adequate variety can be achieved (e.g., use of spices, rotating foods) or others means of warding off menu fatigue. Ultimately, we still need to understand the state of health and performance in relation to the food resource restrictions that must be met for future

	missions.
Bibliography Type:	Description: (Last Updated: 11/22/2025)
Articles in Peer-reviewed Journals	Douglas GL, Bell ST, Roma PG, Oswald T, Young M. "Food acceptability and selection by astronauts on International Space Station missions informs strategies and risks for deep space exploration." Front Psychol. 2025 Oct 30;16:1562044. https://doi.org/10.3389/fpsyg.2025.1562044 ; PMID: 41245329; PMCID: PMC12611640 , Oct-2025